Sun , 21 Sept 2025
Bootcamp of Mathematics (BOOM) 2025

Preprocessing Data

The Foundation of Reliable Machine Learning
By Muhammad Nanda S. / Data Scientist

Session Objectives

01. Understand **the critical importance of data preprocessing** in the Machine Learning pipeline.

02. Identify **common data quality issues** like missing values, outliers, and categorical data.

03. Learn **fundamental techniques to clean and prepare data** for modeling.

04. Recognize that **data preprocessing is an iterative, non-linear process** that **requires critical thinking.**

What to do first?

*after we have the data

Rename Columns?
Data Cleaning?
Exploratory Data Analysis?
Data Visualization?
Remove Missing Value?
Data Splitting?
Or what..?

Data Understanding

Data Understanding, How?

- About data:
  - Data description
  - Metadata
- Basic (and quick) Inspection:
  - data info: data size, data type, missing value, etc.
- Exploratory Data Analysis (EDA):
  - Data Visualization
  - Statistical Descriptive

Further Process

- Data Cleaning:
  - Fixing data type and formatting
  - Remove missing value
  - Remove duplicate data
  - etc.

Why is it so important?

- Wastes computational resources.
- Leads to inaccurate models
- Causes biased predictions

What does "Dirty Data" look like?

Name	Semester	GPA	Algebra Test Score	Class Attendance (%)	Active Organization?
Jefry	5	3.51	85	95	yes
Syahrul	Five	3.2	75	105	no
Null	4	3.75	25	92	y
Izzuddin	6	Three point eight	91	88	No
Yesaya	5.0	2,9		80	YES
Jefry	5	3.51	85	95	yes
Dion	‘5’	0	‘90’	70	yes

- Inconsistent Format
- Duplicate
- Missing Value
- Unreasonable Value
- Outlier

Techniques to Clean Data

Handling Missing Values

- Remove
- Impute:
  - Mean/median/mode
  - Interpolate
  - Rolling window

Handling Outlier

- Remove
- Capping
  - Percentile
  - IQR
  - Z-Score

Encoding

- One-Hot Encoding
- Label Encoding

Scaling

- Normalization
- Standarization

Techniques to Clean Data

Handling Missing Values

- Remove
- Impute:
  - Mean/median/mode
  - Interpolate
  - Rolling window

Handling Outlier

- Remove
- Capping
  - Percentile
  - IQR
  - Z-Score

Encoding

- One-Hot Encoding
- Label Encoding

Scaling

- Normalization
- Standarization

Custom (and best): Subject Matter Expert (SME) input

Additional: Other / Further Process

Data Aggregation

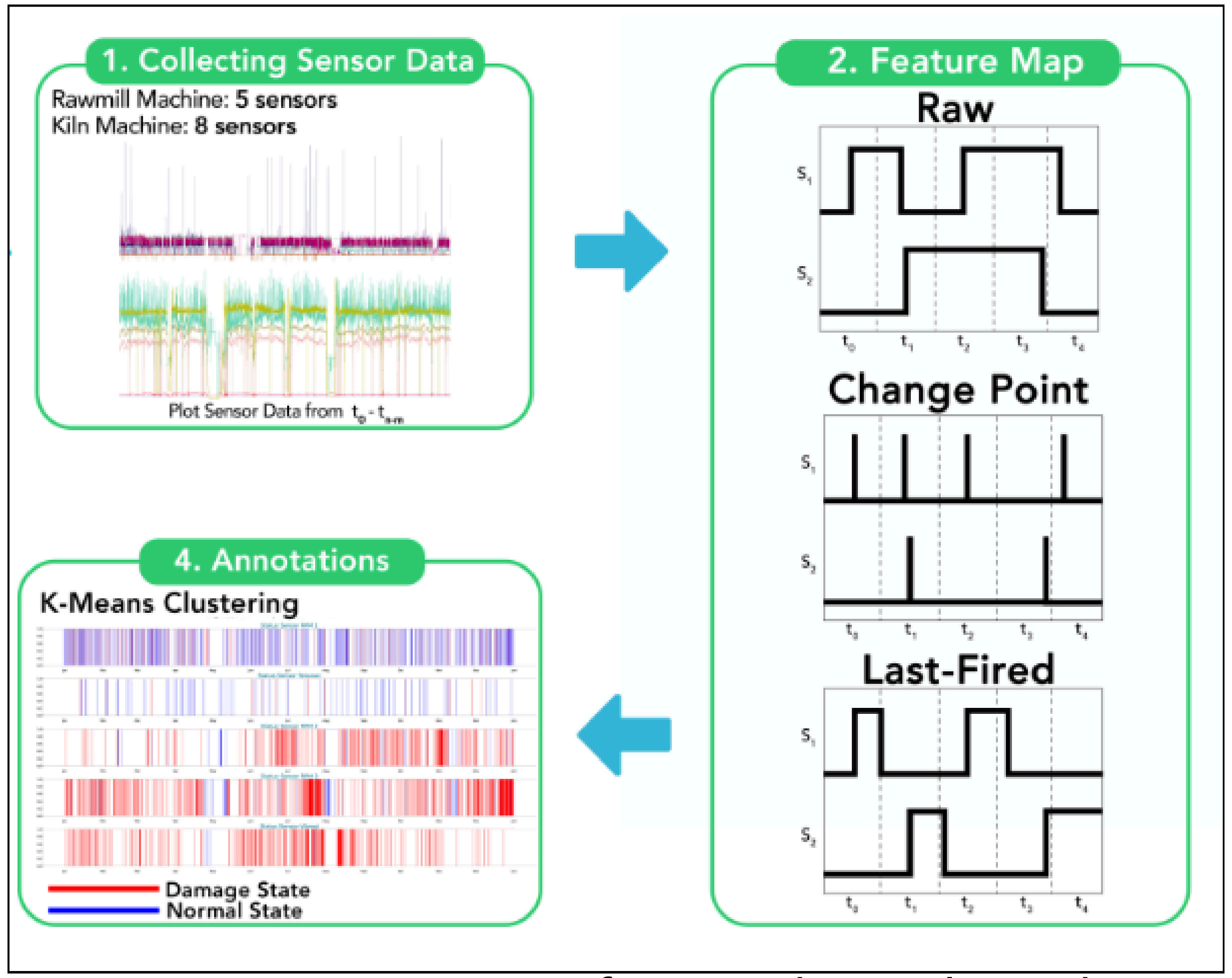
- Merge / concatenate multiple data
- Grouping
- Pivotting

Feature Engineering

- Simple calculation
- Custom calculation

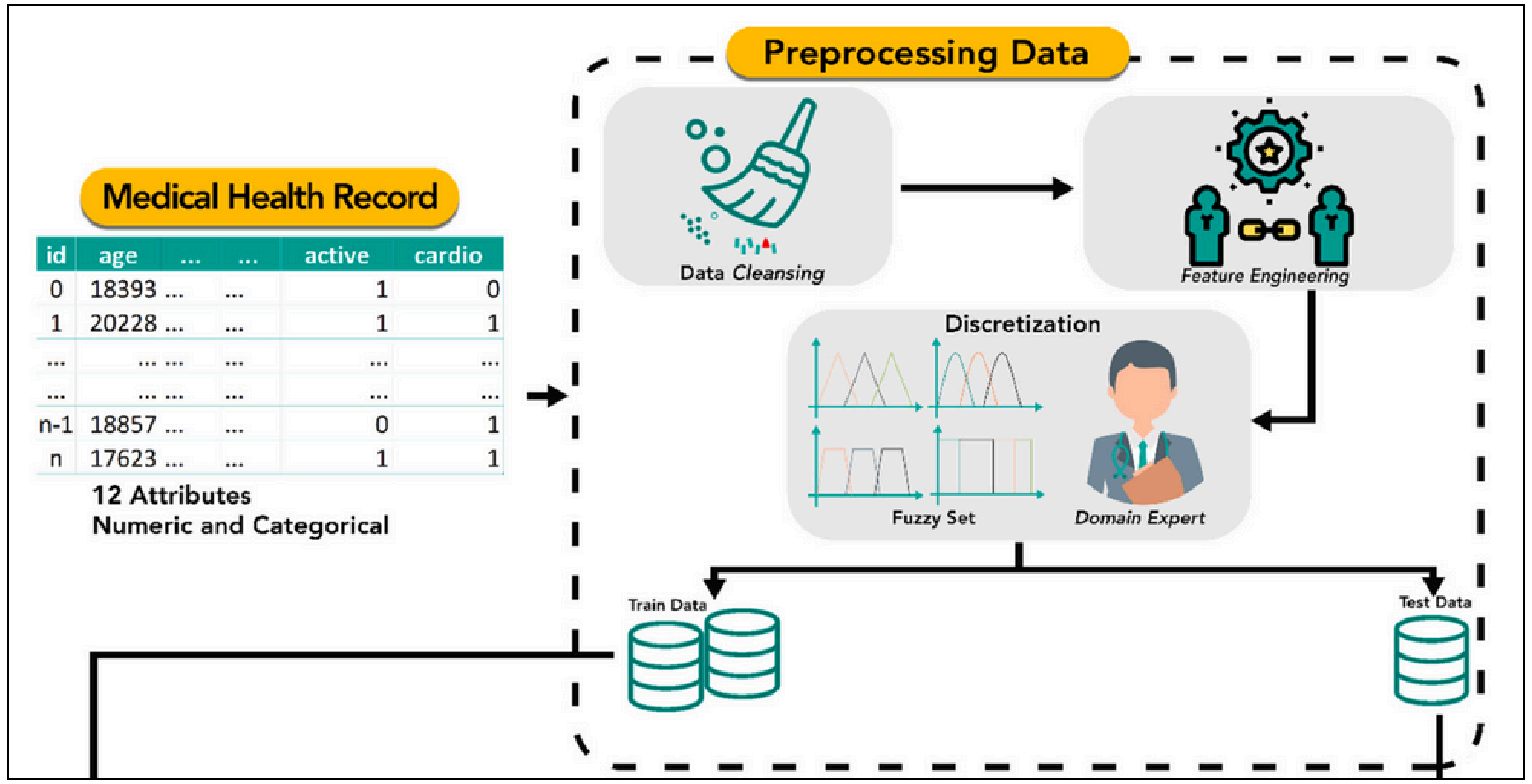
Custom (and best): Subject Matter Expert (SME) input

Data Processing in Action



Use Case: Sensor Data from Industrial Machines

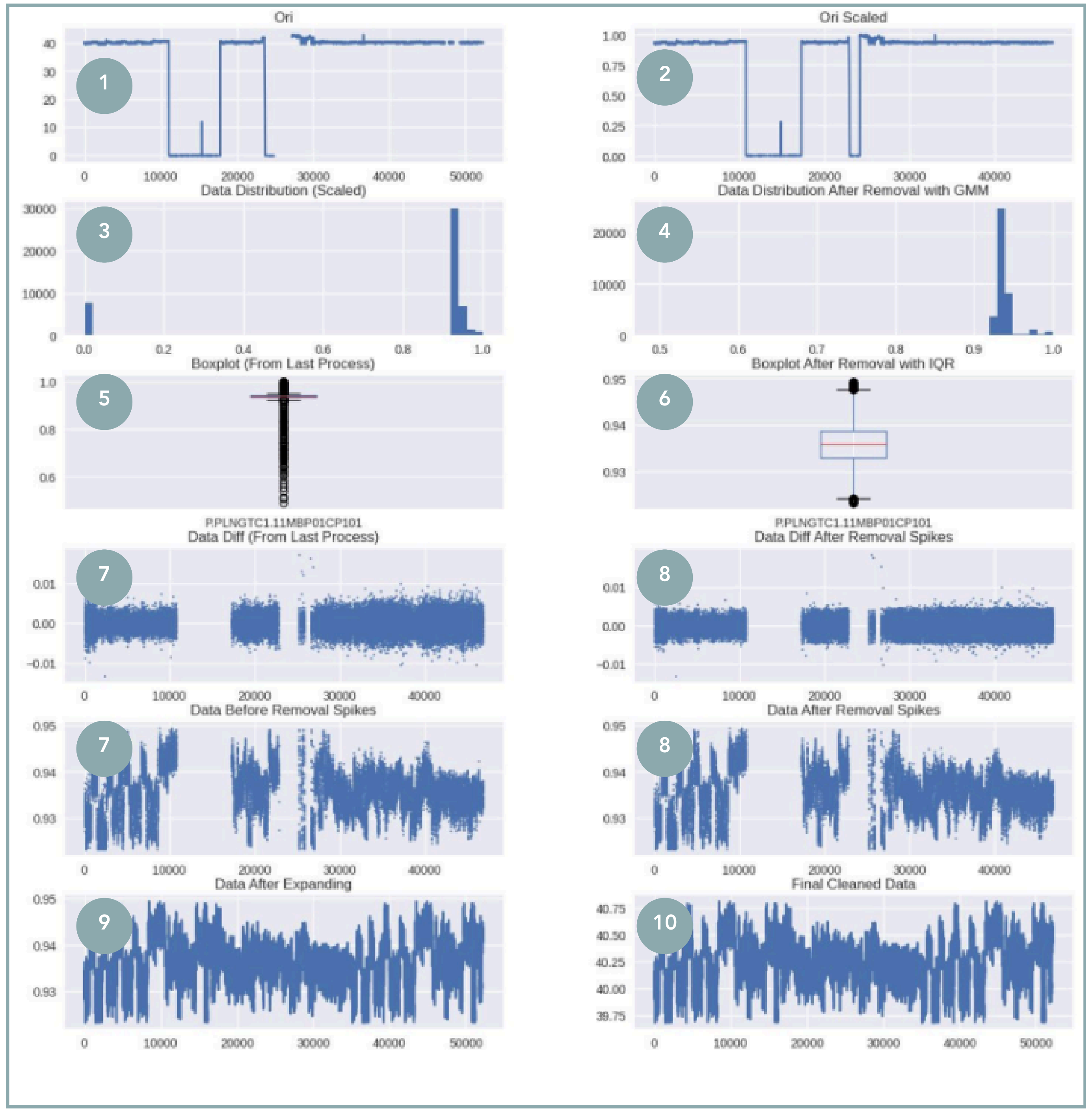
1. Feature Engineering
2. Labeling with Unsupervised Learning



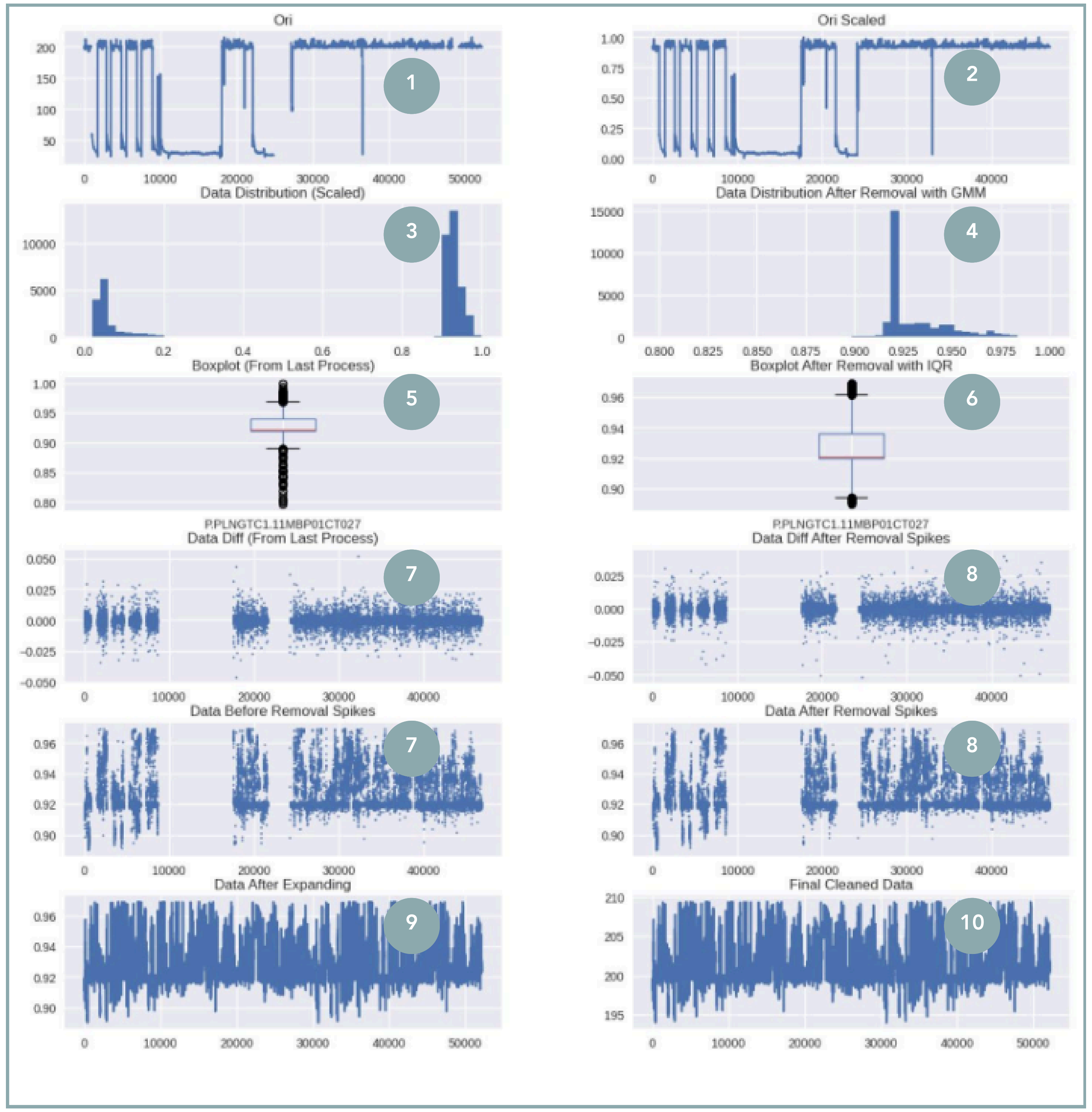
Use Case: Tabular Medical Health Records (Mixed Data Types)

1. Unsupervised Learning
2. Feature Engineering
3. Discretization

Data Processing in Action



Sensor A

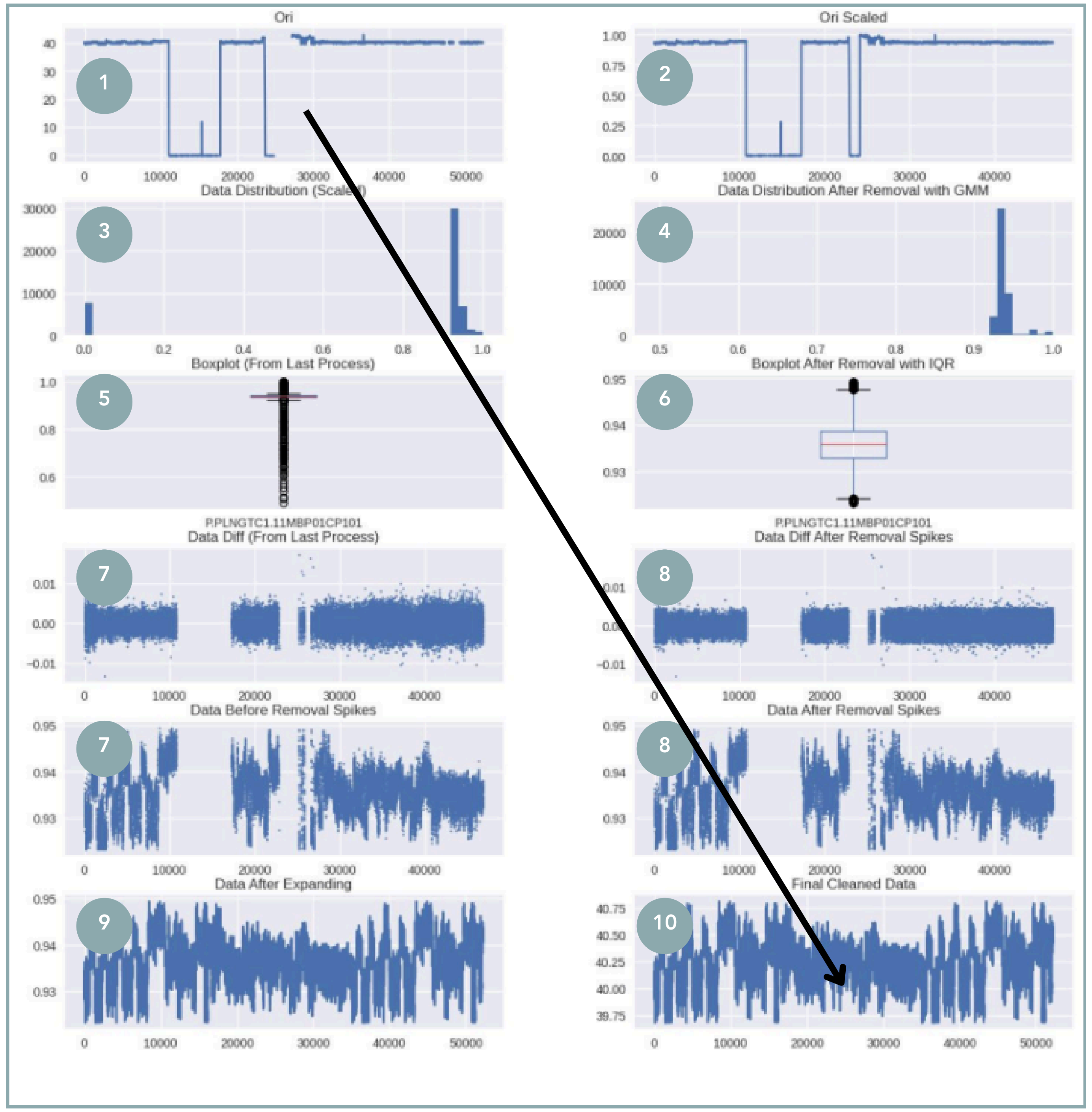


Sensor B

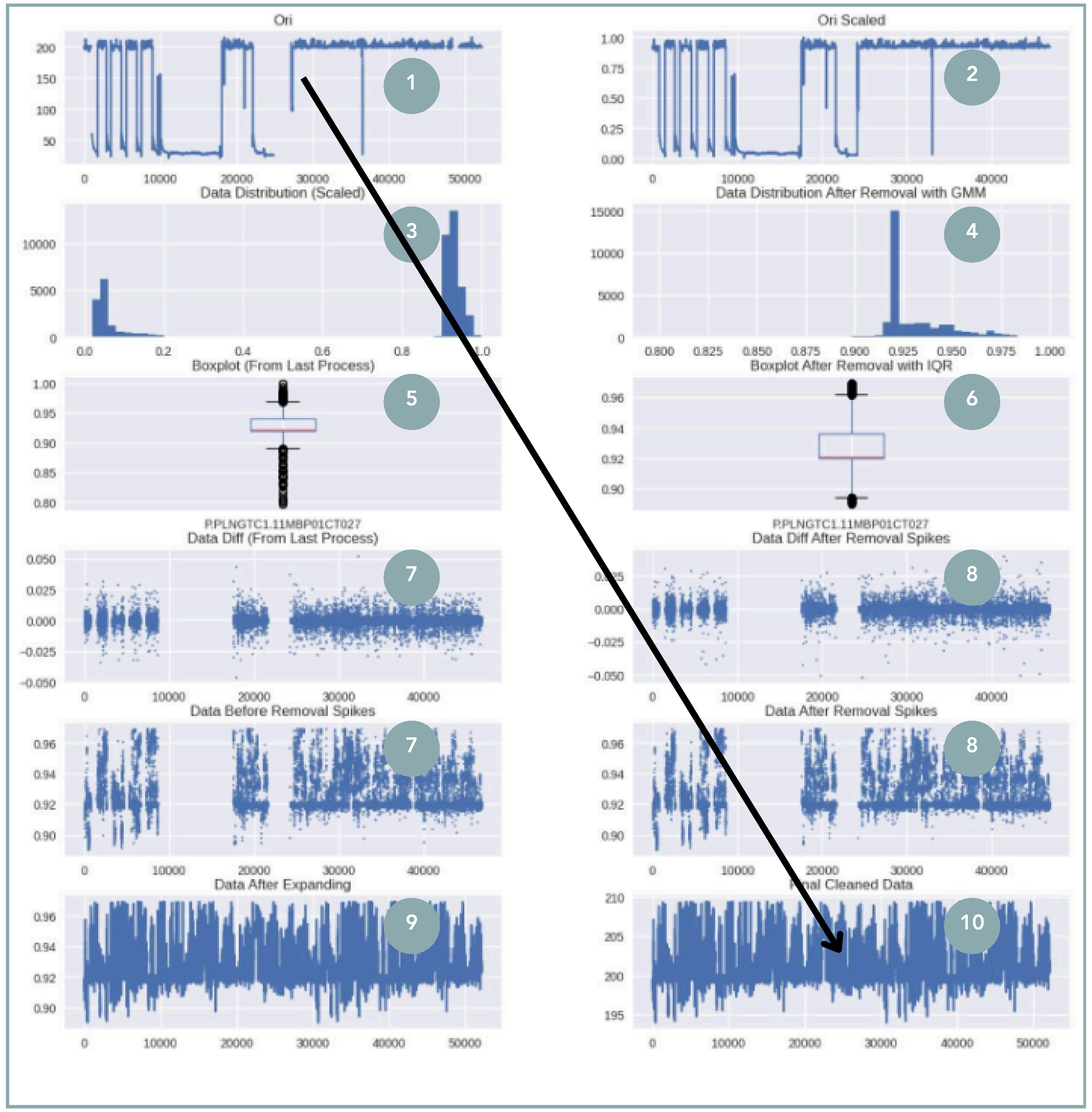
Use Case: Sensor Data from Industrial Machines

1. Scaling
2. Outlier Removal with GMM
3. Outlier Removal with IQR Method
4. Remove Spikes
5. Remove Null & Compacting Data
6. Data Expansion
7. Scale Back

Data Processing in Action



Sensor A

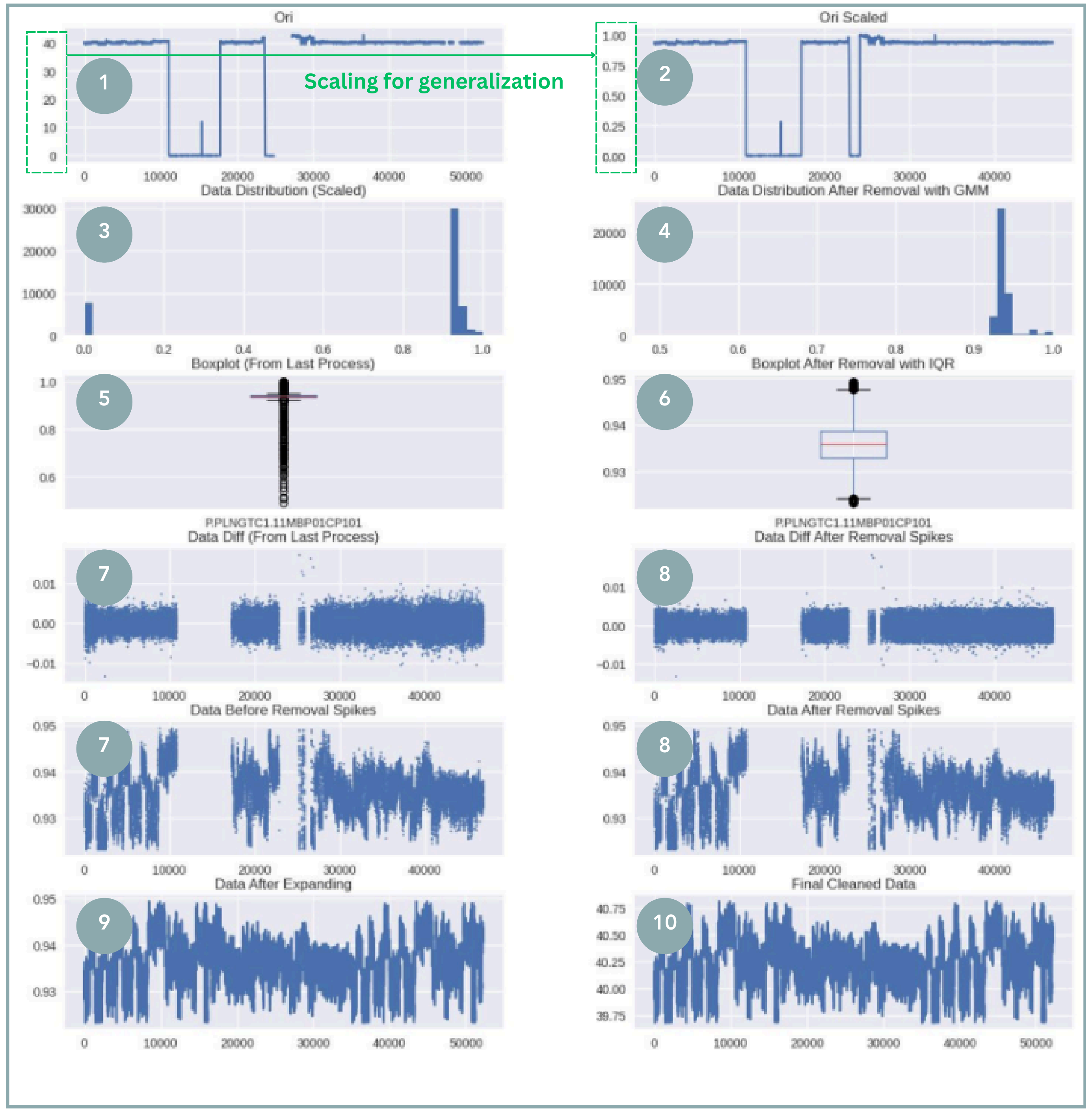


Sensor B

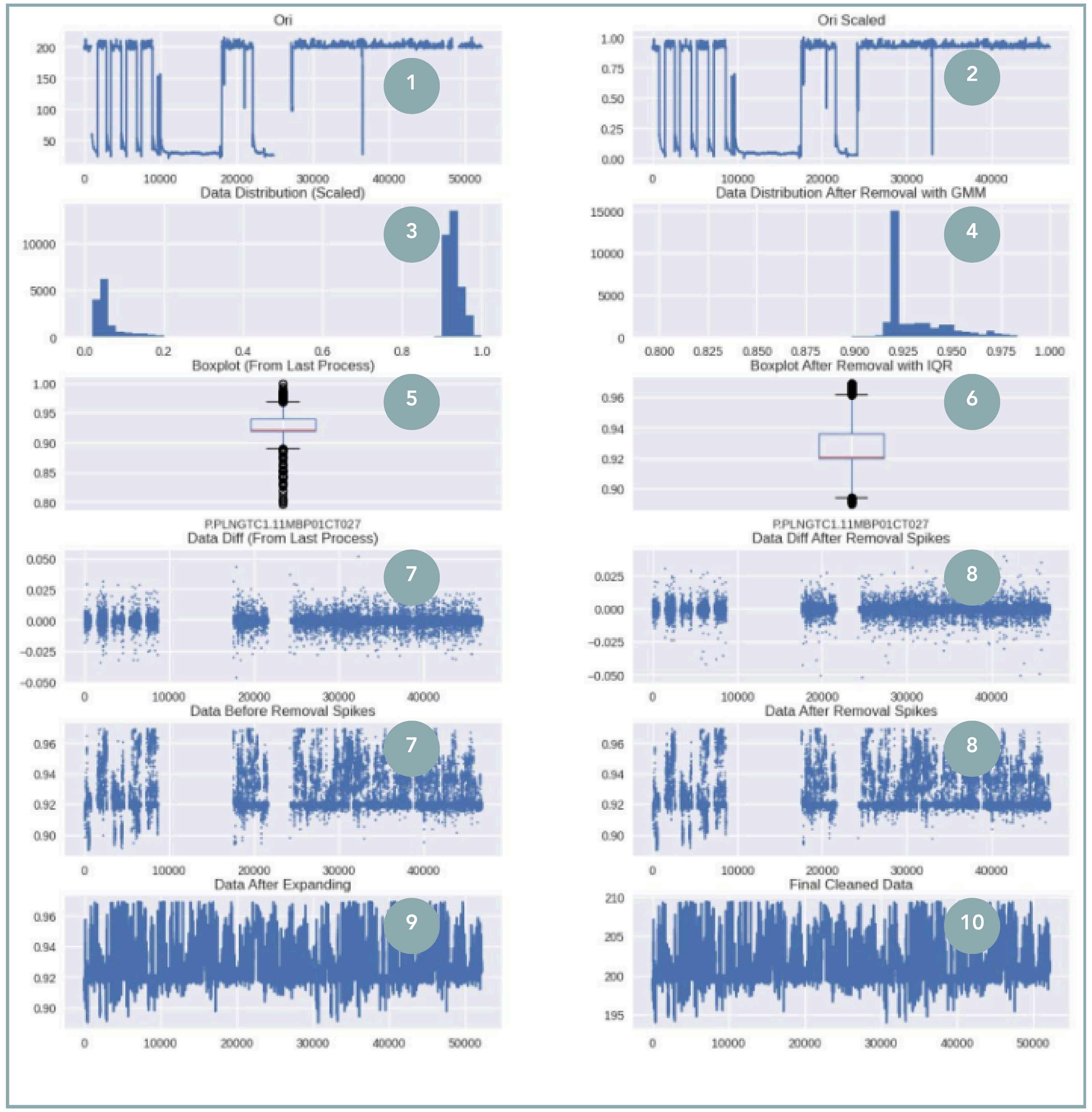
Use Case: Sensor Data from Industrial Machines

1. Scaling
2. Outlier Removal with GMM
3. Outlier Removal with IQR Method
4. Remove Spikes
5. Remove Null & Compacting Data
6. Data Expansion
7. Scale Back

Data Processing in Action



Sensor A

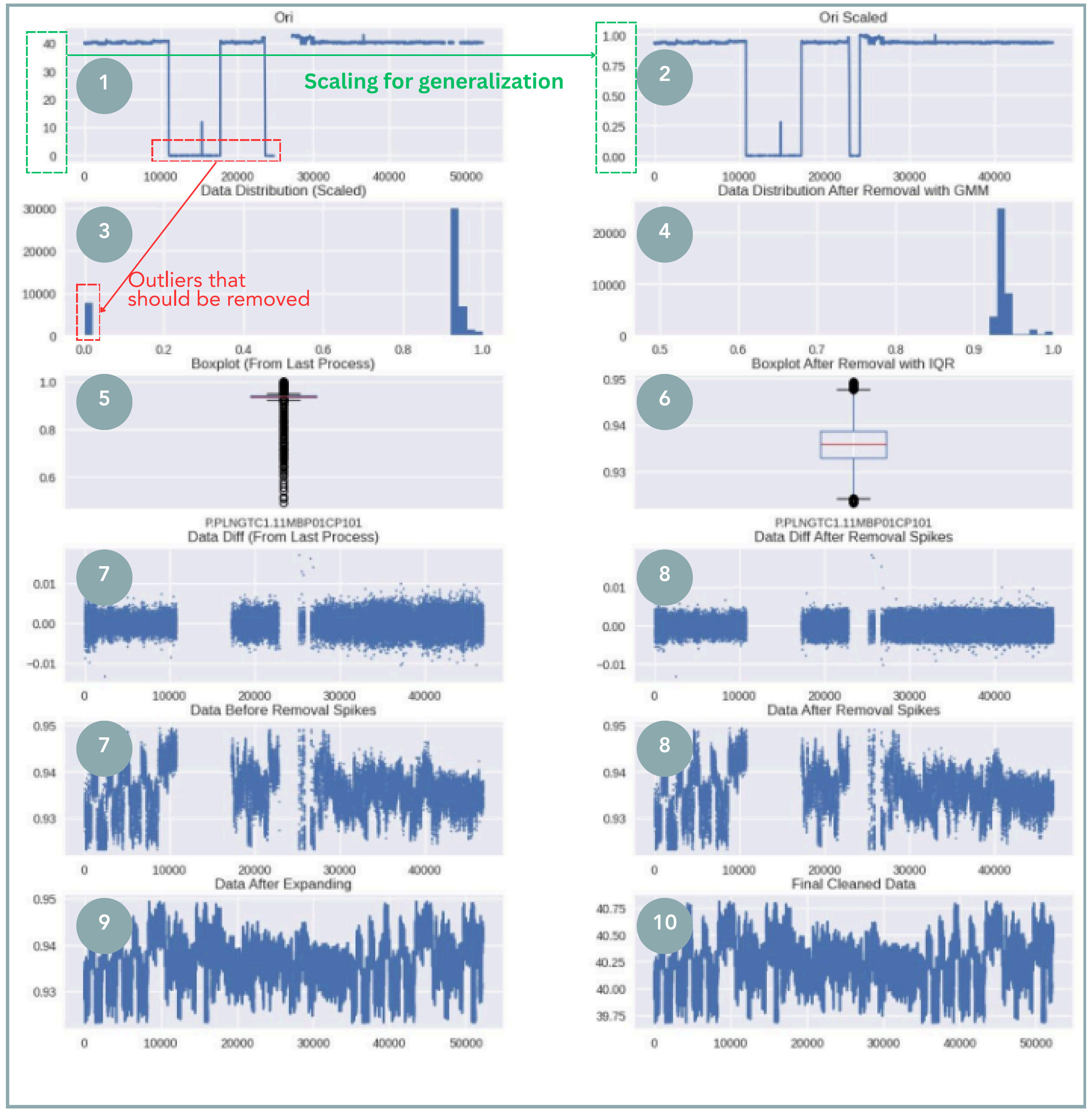


Sensor B

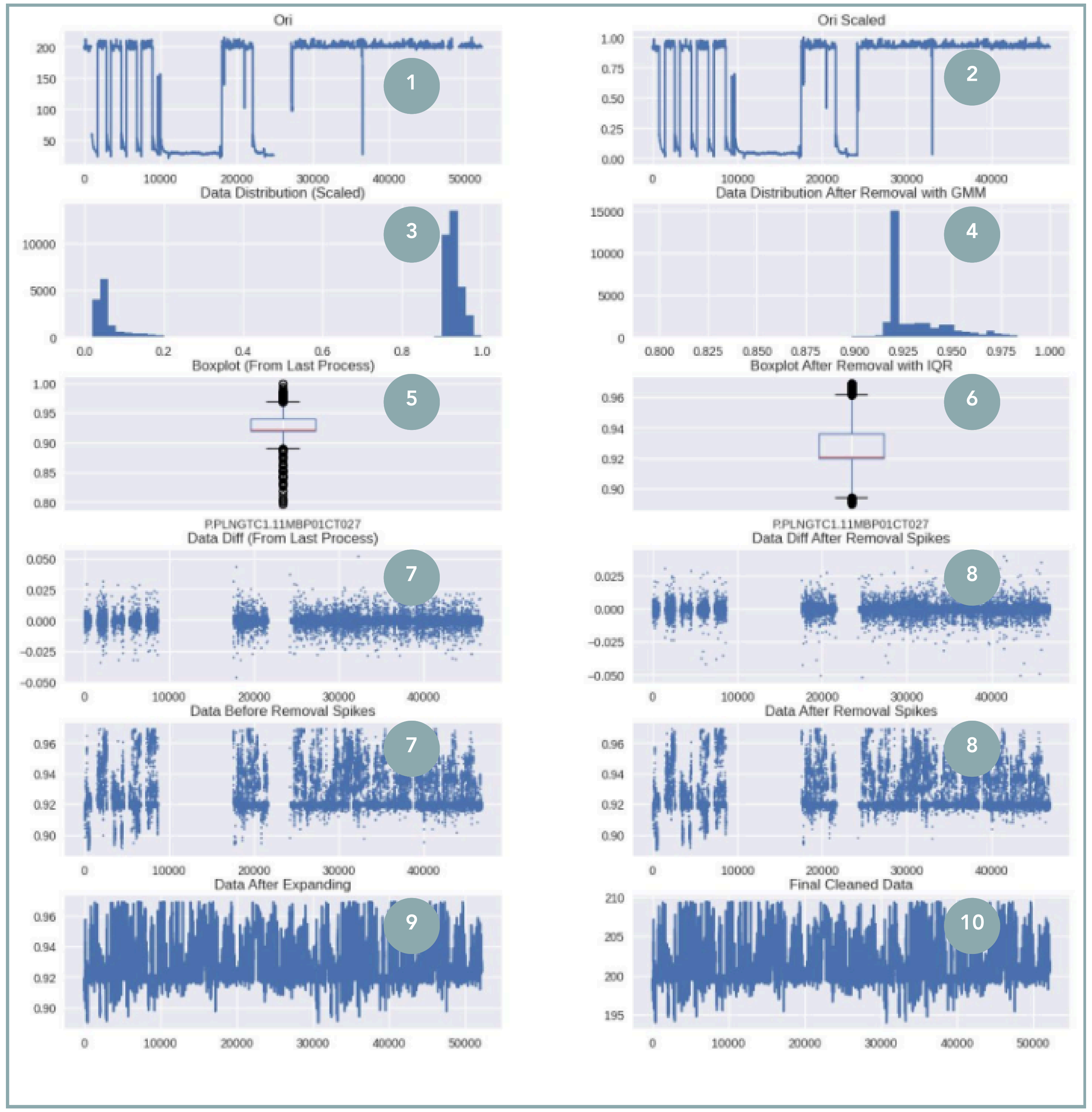
Use Case: Sensor Data from Industrial Machines

1. Scaling
2. Outlier Removal with GMM
3. Outlier Removal with IQR Method
4. Remove Spikes
5. Remove Null & Compacting Data
6. Data Expansion
7. Scale Back

Data Processing in Action



Sensor A

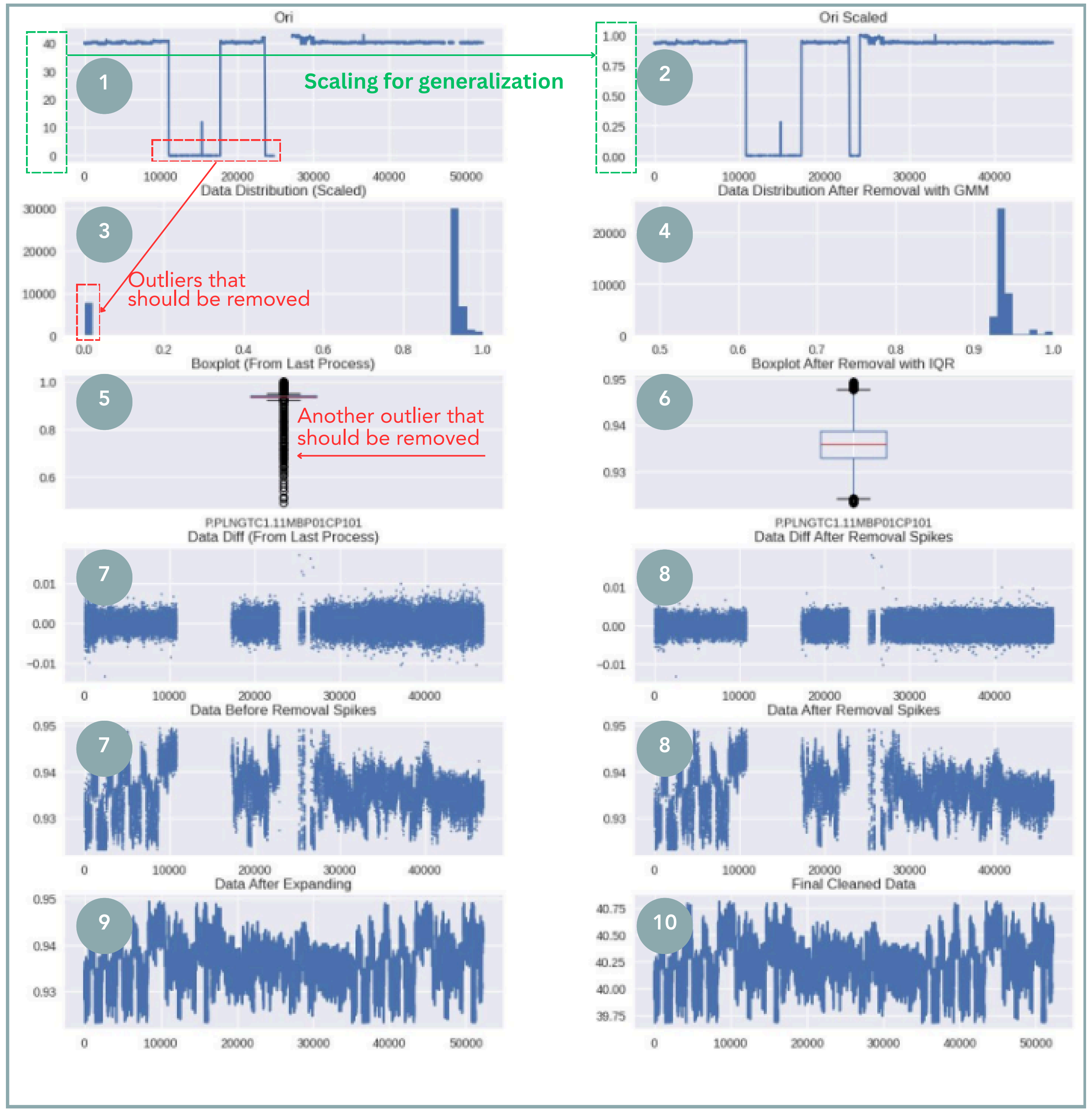


Sensor B

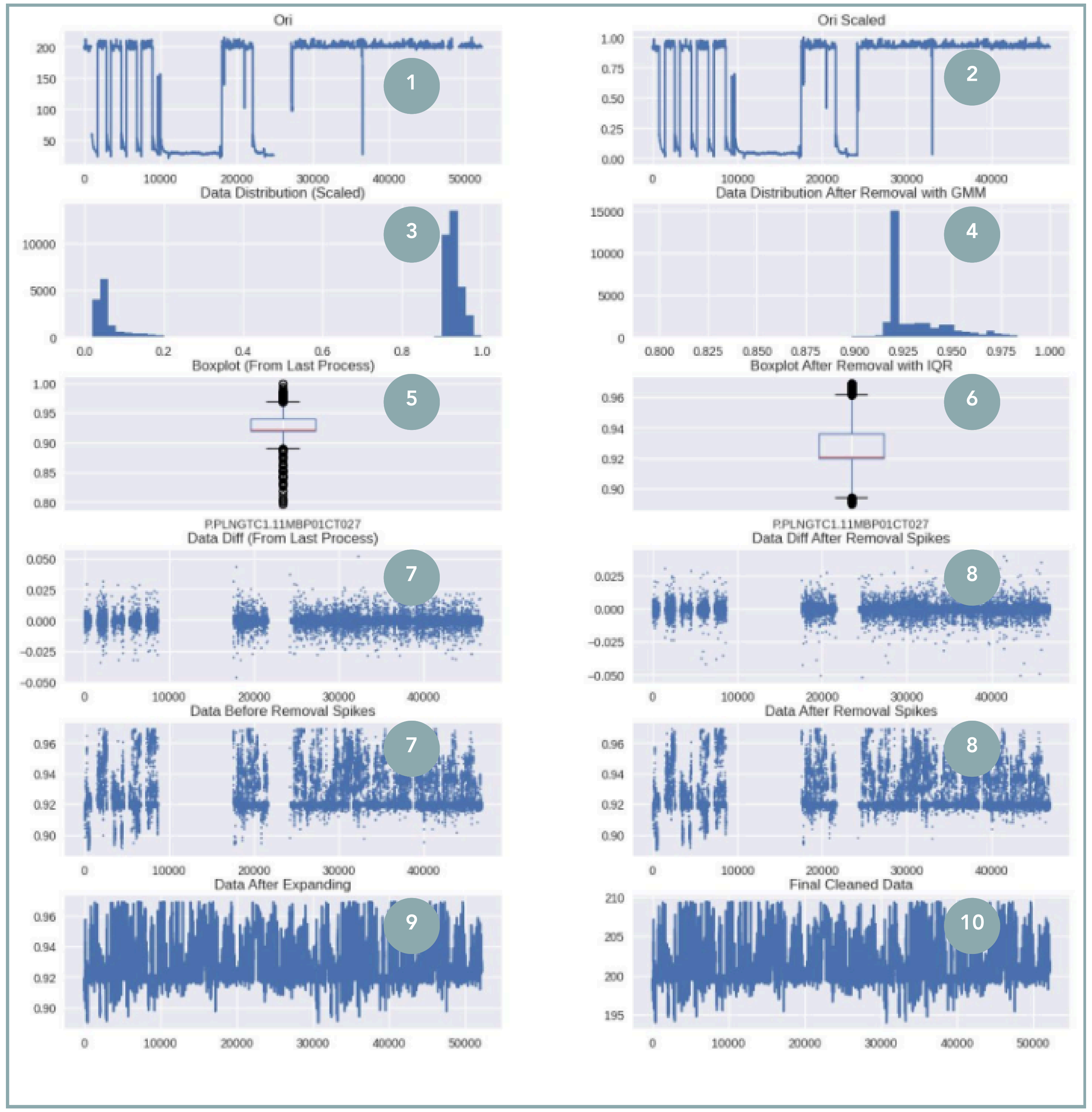
Use Case: Sensor Data from Industrial Machines

1. Scaling
2. Outlier Removal with GMM
3. Outlier Removal with IQR Method
4. Remove Spikes
5. Remove Null & Compacting Data
6. Data Expansion
7. Scale Back

Data Processing in Action



Sensor A

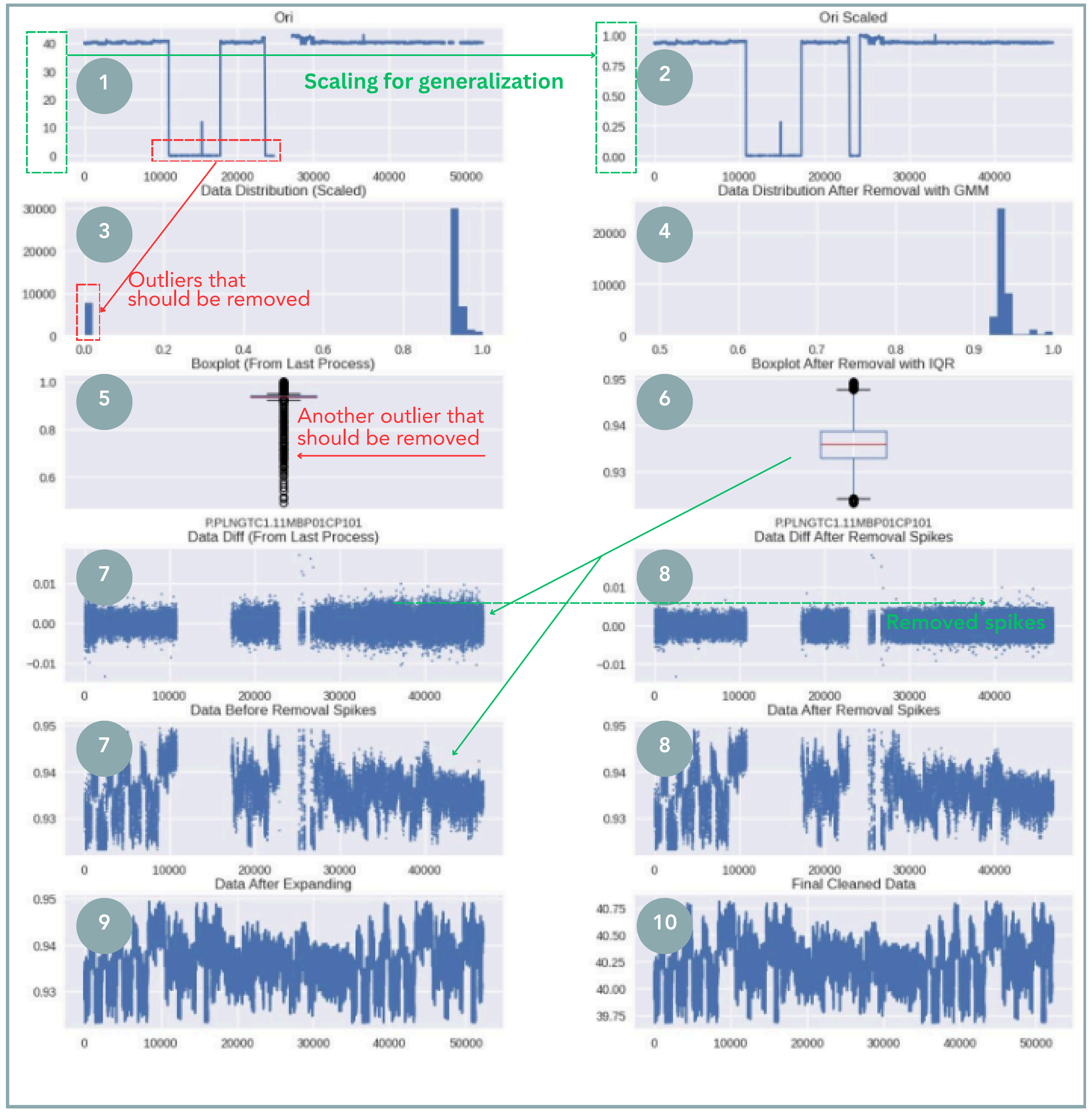


Sensor B

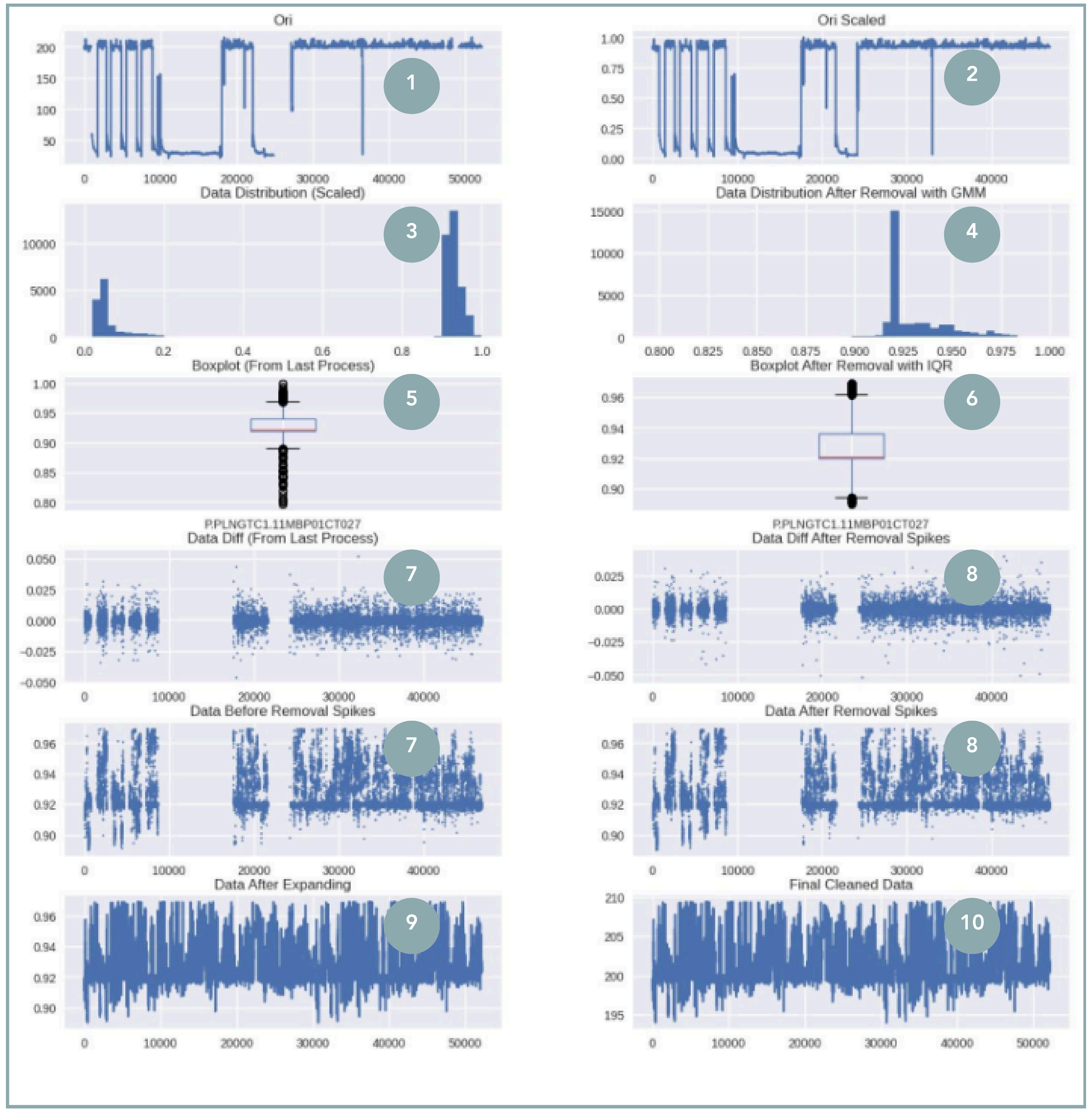
Use Case: Sensor Data from Industrial Machines

1. Scaling
2. Outlier Removal with GMM
3. Outlier Removal with IQR Method
4. Remove Spikes
5. Remove Null & Compacting Data
6. Data Expansion
7. Scale Back

Data Processing in Action



Sensor A

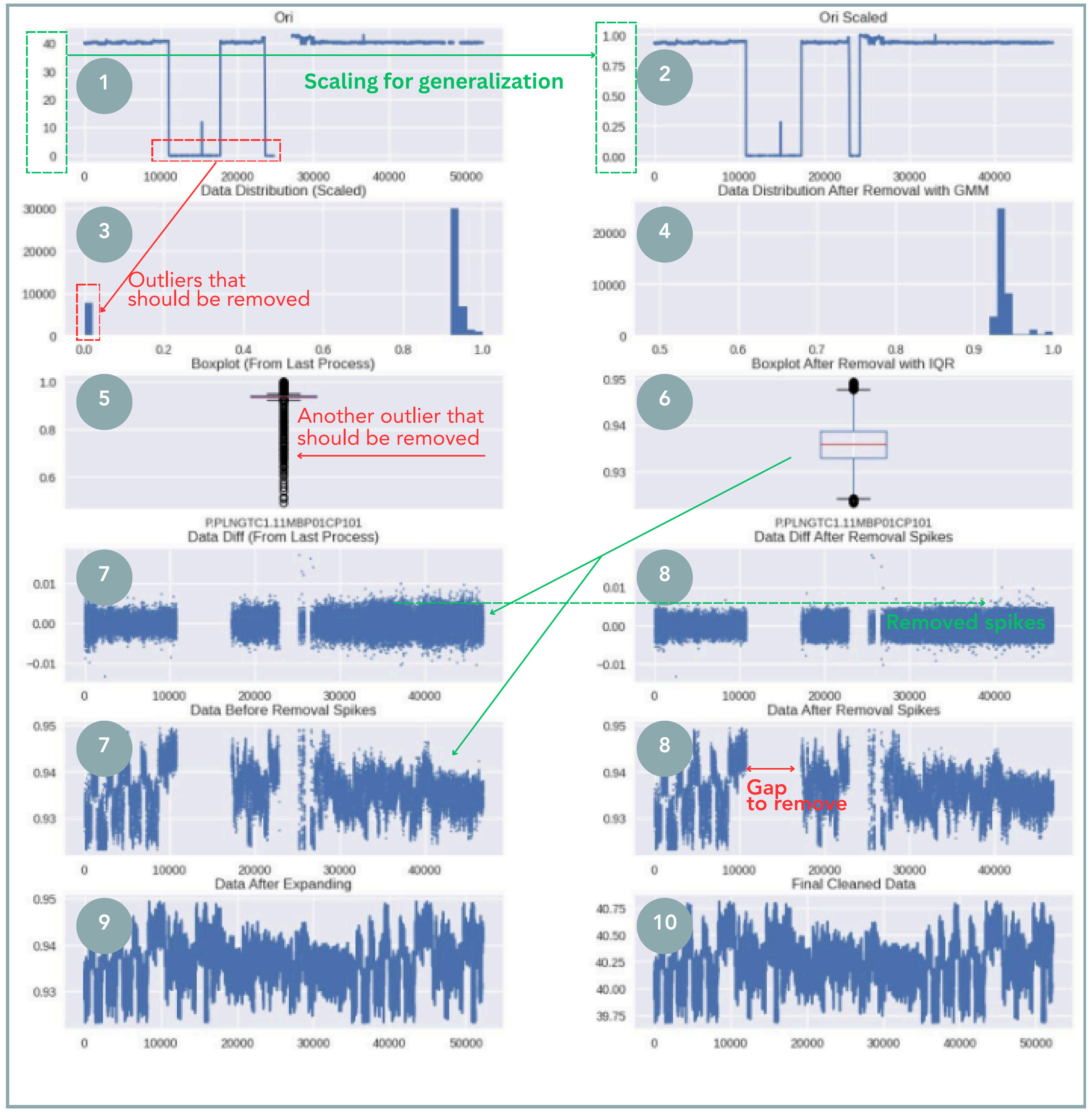


Sensor B

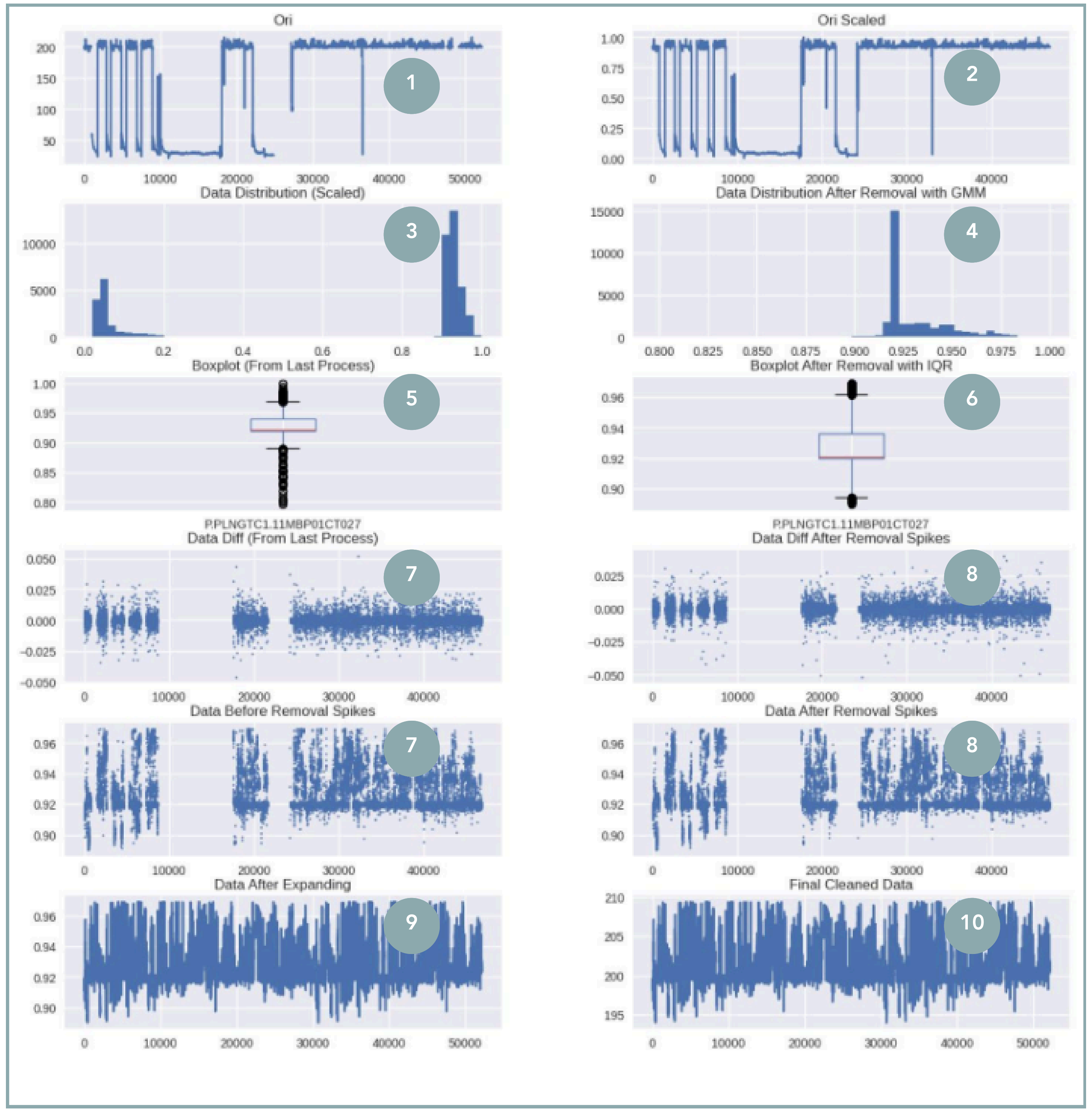
Use Case: Sensor Data from Industrial Machines

1. Scaling
2. Outlier Removal with GMM
3. Outlier Removal with IQR Method
4. Remove Spikes
5. Remove Null & Compacting Data
6. Data Expansion
7. Scale Back

Data Processing in Action



Sensor A



Sensor B

Use Case: Sensor Data from Industrial Machines

1. Scaling
2. Outlier Removal with GMM
3. Outlier Removal with IQR Method
4. Remove Spikes
5. Remove Null & Compacting Data
6. Data Expansion
7. Scale Back


Data Cleaning.....
takes your soul!
meme-arsenal.ru

The Key: Your Soft Skills

Problem Understanding


Curiousity

Critical Thinking


Communication

Let’s Discuss!

Do you have any question?